Deformable DETR: Deformable Transformers for End-To-End Object Detection

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- Problem/Objective
  - Object Detection
- Contribution/Key Idea
  - Propose Deformable DETR
    - Propose Deformable Attention Module
    - Propose Multi-scale Deformable Attention Module
  - Better Performance than DETR
  - 10x Less Training Times

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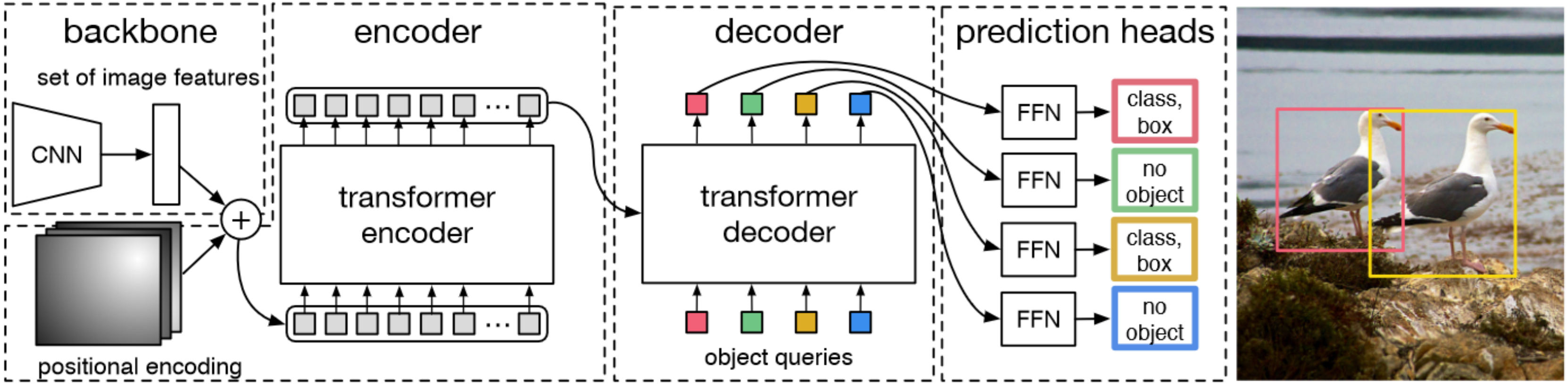
- **DETR (DEtection TRansformer) [1]**



[1] Carion, Nicolas, et al. "End-to-end object detection with transformers." *European conference on computer vision*. Cham: Springer International Publishing, 2020.

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- **DETR Problems**

1) **Requires much longer training epochs to converge**
2) **Low performance at detecting small objects**

Method	Epochs	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	params	FLOPs	Training GPU hours	Inference FPS
Faster R-CNN + FPN	109	42.0	62.1	45.5	26.6	45.4	53.4	42M	180G	380	26
DETR	500	42.0	62.4	44.2	20.5	45.8	61.1	41M	86G	2000	28
DETR-DC5	500	43.3	63.1	45.9	22.5	47.3	61.1	41M	187G	7000	12

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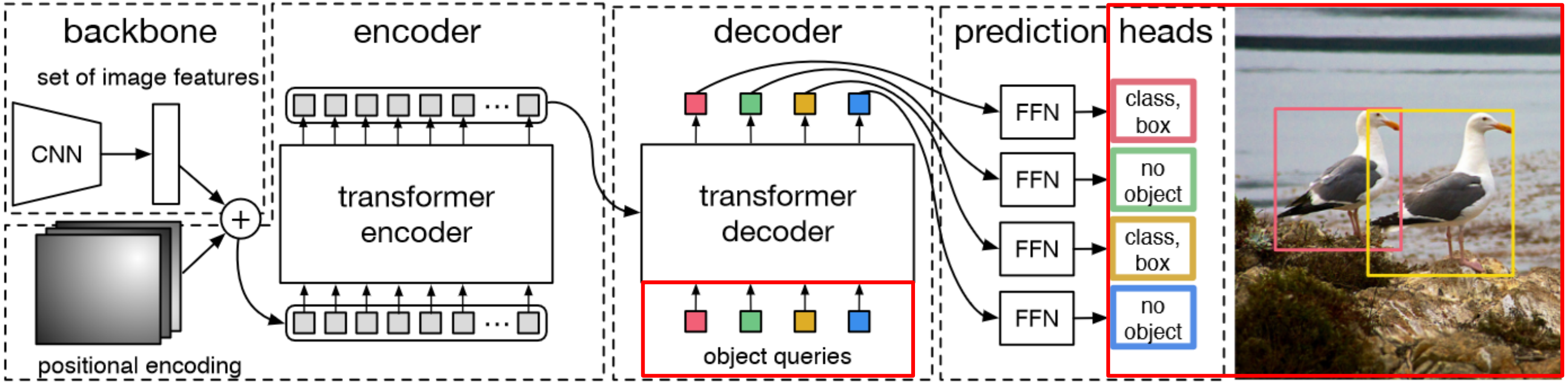
- Cause of DETR Problems

1) Requires much longer training epochs to converge

Bipartite Matching



Random Initialization

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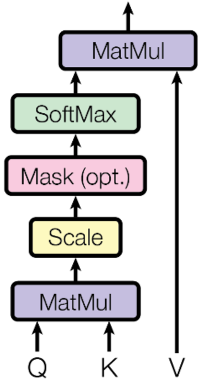
- Cause of DETR Problems

2) Low performance at detecting small objects

Scaled Dot-Product Attention



$$O(H^2W^2C)$$

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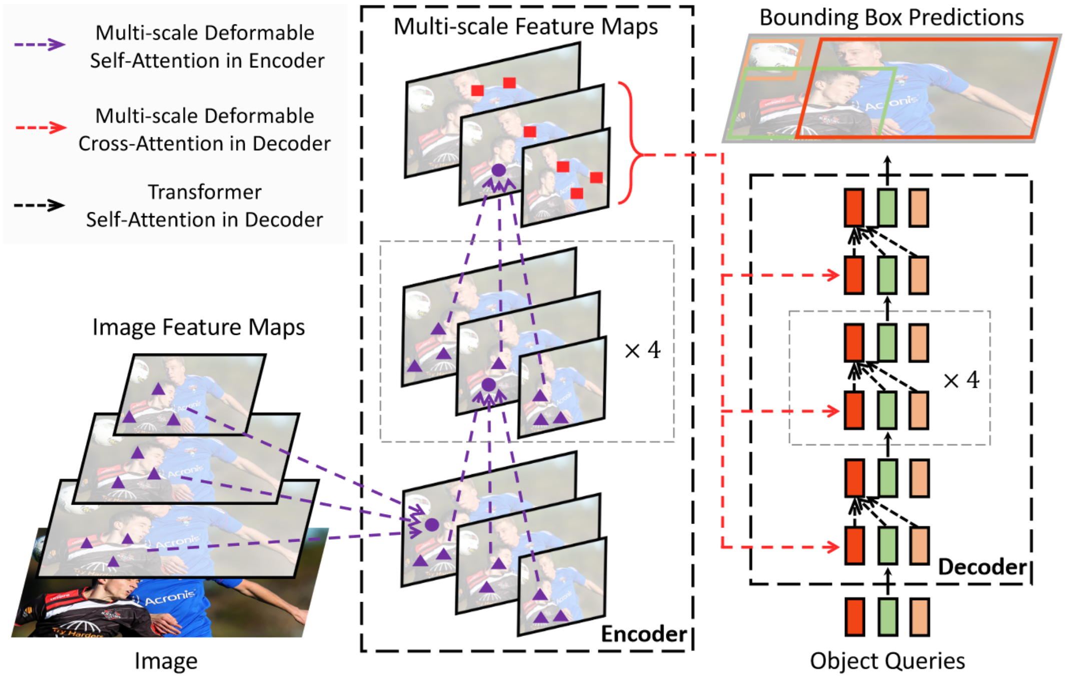
- **Deformable DETR Overview**



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- Notations

Notation	Description
m	index for attention head
l	index for feature level of key element
q	index for query element
k	index for key element
N_q	number of query elements
N_k	number of key elements
M	number of attention heads
L	number of input feature levels
K	number of sampled keys in each feature level for each attention head
C	input feature dimension
C_v	feature dimension at each attention head
H	height of input feature map
W	width of input feature map
H^l	height of input feature map of l^{th} feature level
W^l	width of input feature map of l^{th} feature level
A_{mqk}	attention weight of q^{th} query to k^{th} key at m^{th} head
A_{mlqk}	attention weight of q^{th} query to k^{th} key in l^{th} feature level at m^{th} head
$\boldsymbol{z}_q$	input feature of q^{th} query
$\boldsymbol{p}_q$	2-d coordinate of reference point for q^{th} query
$\hat{\boldsymbol{p}}_q$	normalized 2-d coordinate of reference point for q^{th} query
$\boldsymbol{x}$	input feature map (input feature of key elements)
$\boldsymbol{x}_k$	input feature of k^{th} key
$\boldsymbol{x}^l$	input feature map of l^{th} feature level
$\Delta\boldsymbol{p}_{mqk}$	sampling offset of q^{th} query to k^{th} key at m^{th} head
$\Delta\boldsymbol{p}_{mlqk}$	sampling offset of q^{th} query to k^{th} key in l^{th} feature level at m^{th} head
$\boldsymbol{W}_m$	output projection matrix at m^{th} head
$\boldsymbol{U}_m$	input query projection matrix at m^{th} head
$\boldsymbol{V}_m$	input key projection matrix at m^{th} head
$\boldsymbol{W}'_m$	input value projection matrix at m^{th} head
$\phi_l(\hat{\boldsymbol{p}})$	unnormalized 2-d coordinate of $\hat{\boldsymbol{p}}$ in l^{th} feature level
exp	exponential function
σ	sigmoid function
σ^{-1}	inverse sigmoid function

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- **Deformable Attention Module**

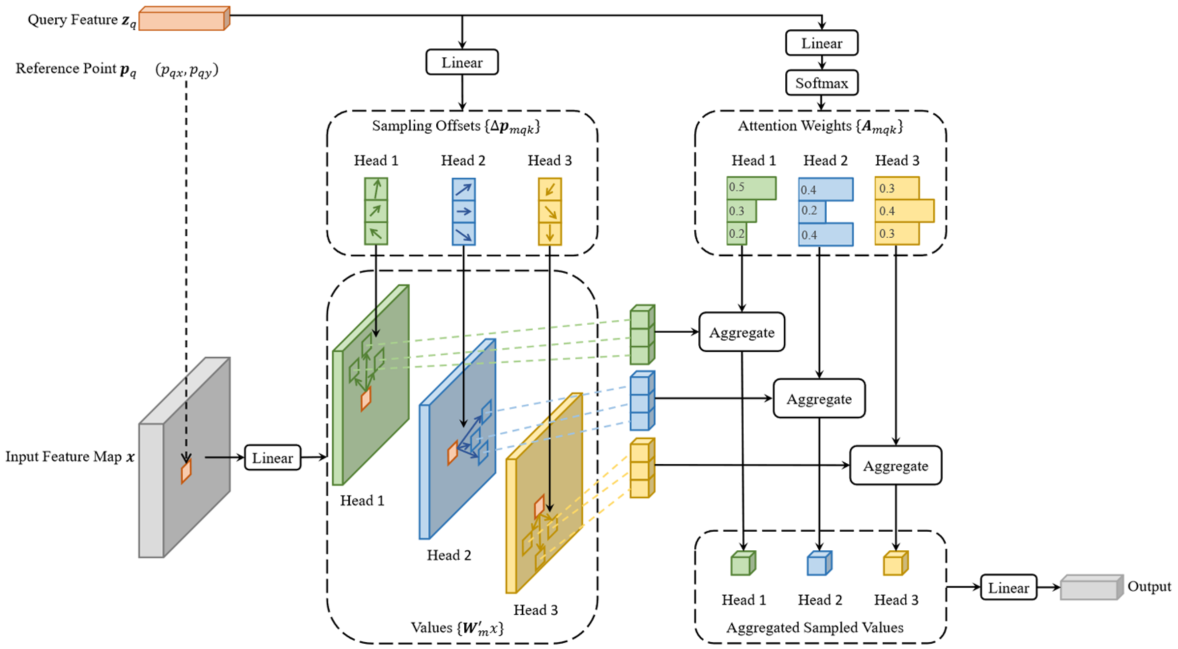

Figure 2: Illustration of the proposed deformable attention module.

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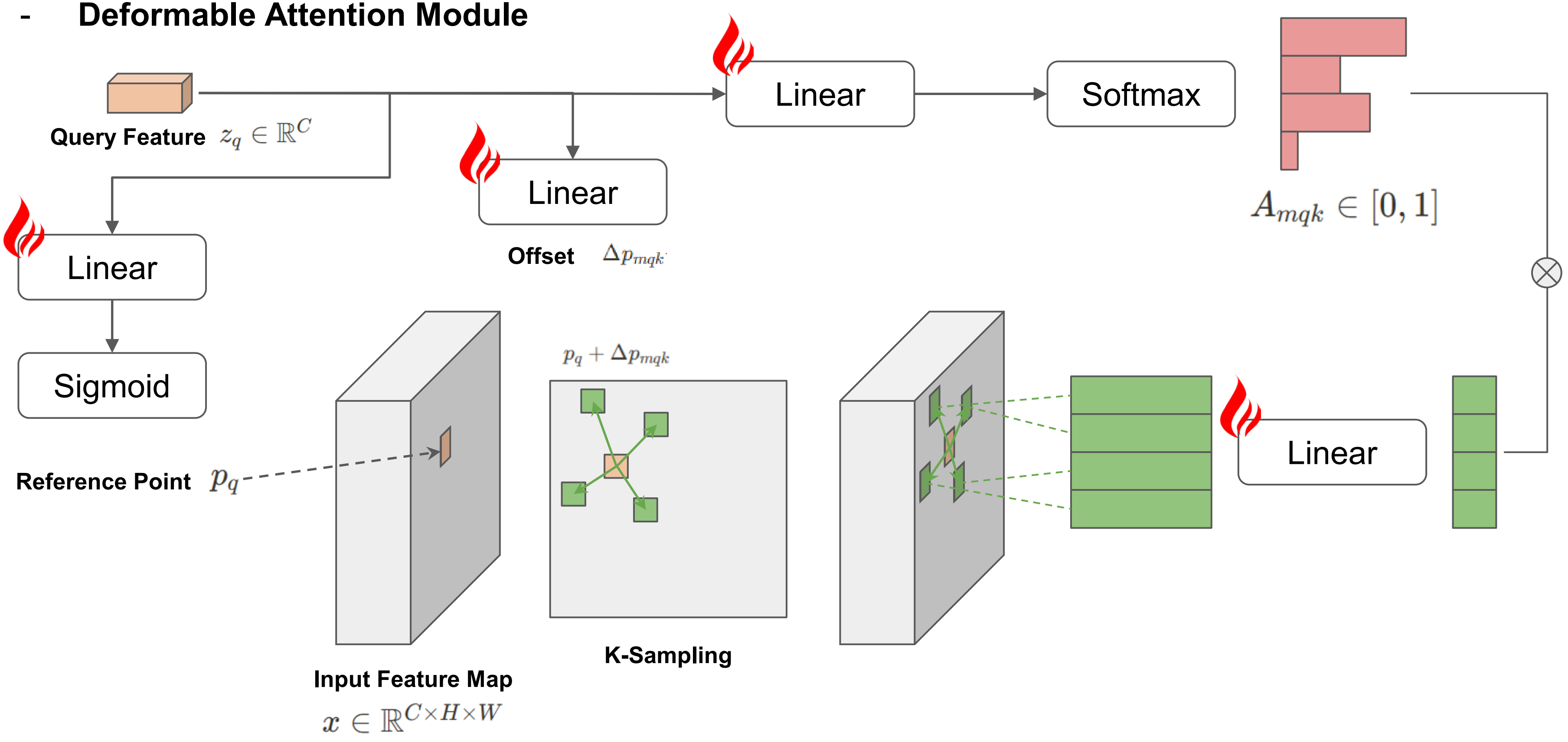
- **Deformable Attention Module**



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- Deformable Attention Module

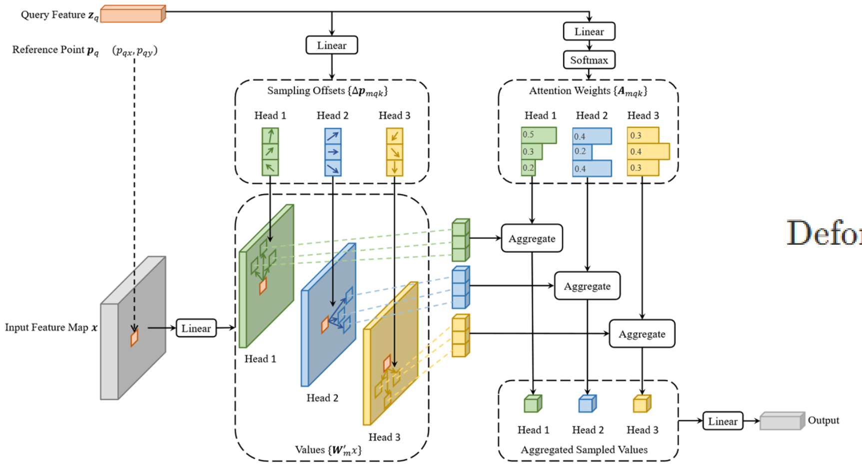

Figure 2: Illustration of the proposed deformable attention module.

$$\mathrm{DeformAttn}(z_q, p_q, x) = \sum_{m=1}^{M} W_m \left[\sum_{k=1}^{K} A_{mqk} \cdot W'_m x(p_q + \Delta p_{mqk}) \right]$$

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- **Deformable Attention Pros**

1. **모든 픽셀을 계산하지 않고 소수의 샘플링 위치만 선택해서 계산량이 적음**
2. **Query가 중요한 샘플링 위치에만 집중하므로 학습 시 수렴 속도가 빠름**
3. **Reference Point를 기준으로 offset 학습을 하기 때문에, 다양한 객체의 크기와 위치를 효과적으로 처리할 수 있음**

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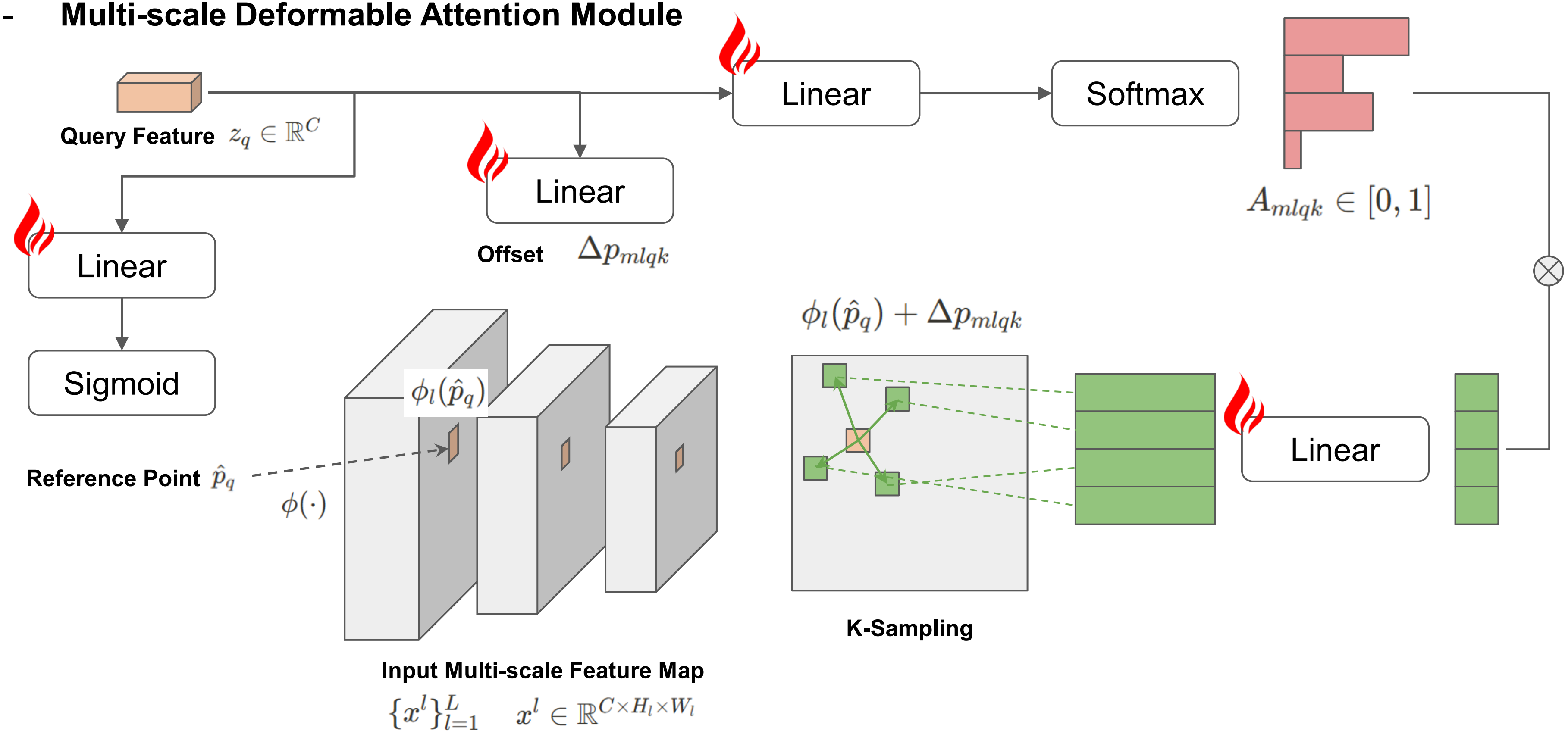
- Multi-scale Deformable Attention Module



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- **Multi-scale Deformable Attention Module**

$$\text{MSDeformAttn}(z_q, \hat{p}_q, \{x^l\}_{l=1}^{L}) = \sum_{m=1}^{M} W_m \left[\sum_{l=1}^{L} \sum_{k=1}^{K} A_{mlqk} \cdot W'_m x^l(\phi_l(\hat{p}_q) + \Delta p_{mlqk}) \right]$$

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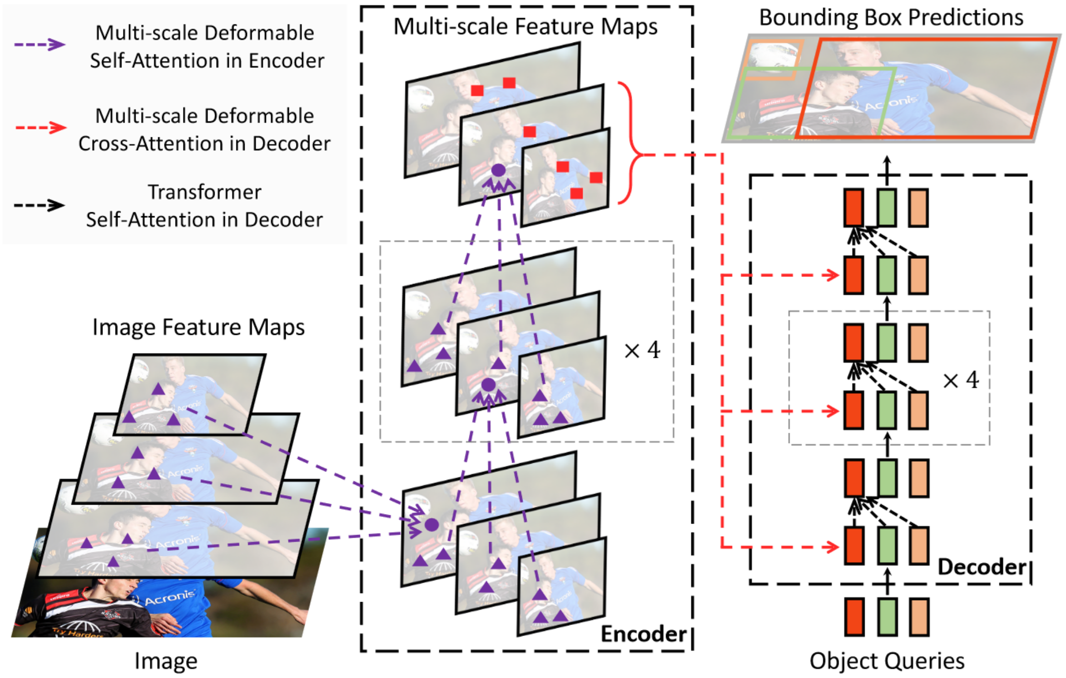
- **Deformable DETR**



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- **Experiments**

Table 1: Comparision of Deformable DETR with DETR on COCO 2017 val set. DETR-DC5$^+$ denotes DETR-DC5 with Focal Loss and 300 object queries.

Method	Epochs	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	params	FLOPs	Training GPU hours	Inference FPS
Faster R-CNN + FPN	109	42.0	62.1	45.5	26.6	45.4	53.4	42M	180G	380	26
DETR	500	42.0	62.4	44.2	20.5	45.8	61.1	41M	86G	2000	28
DETR-DC5	500	43.3	63.1	45.9	22.5	47.3	61.1	41M	187G	7000	12
DETR-DC5	50	35.3	55.7	36.8	15.2	37.5	53.6	41M	187G	700	12
DETR-DC5$^+$	50	36.2	57.0	37.4	16.3	39.2	53.9	41M	187G	700	12
Deformable DETR	50	43.8	62.6	47.7	26.4	47.1	58.0	40M	173G	325	19
+ iterative bounding box refinement	50	45.4	64.7	49.0	26.8	48.3	61.7	40M	173G	325	19
++ two-stage Deformable DETR	50	46.2	65.2	50.0	28.8	49.2	61.7	40M	173G	340	19

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- **Experiments**

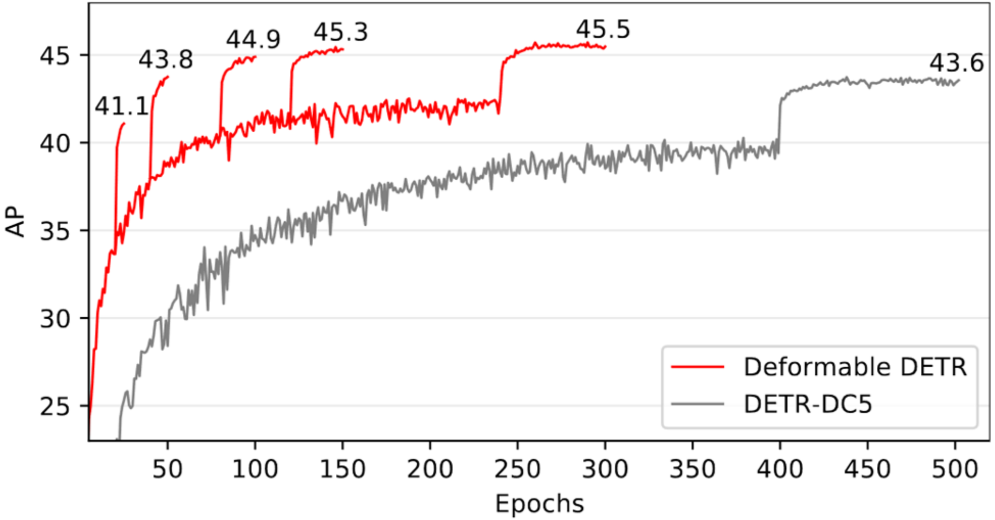

Figure 3: Convergence curves of Deformable DETR and DETR-DC5 on COCO 2017 val set. For Deformable DETR, we explore different training schedules by varying the epochs at which the learning rate is reduced (where the AP score leaps).

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- **Experiments**

Table 2: Ablations for deformable attention on COCO 2017 val set. "MS inputs" indicates using multi-scale inputs. "MS attention" indicates using multi-scale deformable attention. K is the number of sampling points for each attention head on each feature level.

MS inputs	MS attention	K	FPNs	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
✓	✓	4	FPN (Lin et al., 2017a)	43.8	62.6	47.8	26.5	47.3	58.1
✓	✓	4	BiFPN (Tan et al., 2020)	43.9	62.5	47.7	25.6	47.4	57.7
		1	w/o	39.7	60.1	42.4	21.2	44.3	56.0
✓		1		41.4	60.9	44.9	24.1	44.6	56.1
✓		4		42.3	61.4	46.0	24.8	45.1	56.3
✓	✓	4		43.8	62.6	47.7	26.4	47.1	58.0

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- **Visualization**

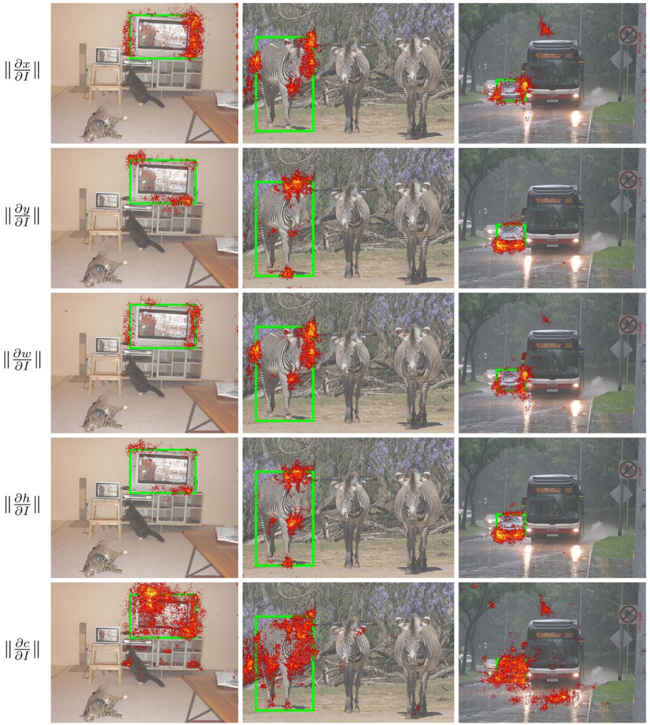

Figure 5: The gradient norm of each item (coordinate of object center (x, y), width/height of object bounding box w/h, category score c of this object) in final detection result with respect to each pixel in input image I.